Project Proposal

Believer Long Range Observation Fixed Wing Drone

Abstract

This project aims to develop a long-range fixed-wing unmanned aerial system for beyond visual line of sight (BVLOS) observation and advanced onboard decision making. Building on an existing Believer airframe prototype developed by the QUT Aerospace Society (QUTAS), the project focuses on upgrading the avionics, navigation, and communications systems to create a robust and scalable research platform. Intended applications include shark spotting, environmental monitoring, and agricultural observation, with integration of image processing and machine learning to support teaching and research activities. Delivered through a student-led, industry-engaged framework, the project provides practical experience in autonomy, sensor fusion, and long-range UAV operations. A total budget of approximately **$1,030** is sought to enable these upgrades and progress the platform toward its full operational capability.

Date: 25/01/2026

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Description:

This project aims to develop a fixed-wing drone for beyond visual line of sight (BVLOS) observation missions. The aim of this project is to have a sophisticated long-range system capable of advanced onboard decision making, including object detection for a diverse range of applications with a particular interest in shark spotting, monitoring of threatened ecosystems, and in agricultural applications.

QUT Aerospace Society (QUTAS) has already commenced work on this project and developed an early prototype that is nearing readiness for its maiden flight. The current system configuration is based on a Believer airframe running ArduPlane on a Matek F405-Wing flight controller. The aircraft communicates with the ground station via a 900 MHz RF link provided by an RFD900 ultra-long-range modem pair, with the ground station implemented on a laptop running QGroundControl. At present, the aircraft can be controlled through QGroundControl by interfacing an RC transmitter as a joystick input (Figure 1).

For navigation and localisation, the system is equipped with a u-blox NEO-7 GNSS receiver, providing access to both GPS and Japan's Quasi-Zenith Satellite System (QZSS). This enables the aircraft to estimate latitude and longitude, altitude, and ground speed, forming the basis for both the onboard filter responsible for stabilisation as well as the basis for situational awareness and future autonomous flight and sensing capabilities.

Contact has been established with QUT's Chief Remote Pilot, the Brisbane North Model Aero Club (BNEMAC), and CSIRO, and initial discussions have commenced regarding suitable locations for safe flight testing and the regulatory frameworks governing such operations. It is hoped that the system will have its maiden flight by the beginning of the first university semester.

Figure 1: "Believer" Airframe and Ground Station

Reasoning & Objectives

The objective of this project is to develop a highly capable remote sensing platform suitable for beyond visual line of sight (BVLOS) operations across a diverse range of mission profiles. As a university recognised nationally for its leadership in robotics and autonomous systems, and in light of the growing role of unmanned aircraft as viable mission platforms, the project aims to demonstrate QUT's technical capability while enhancing the educational experience of its students.

The long-range fixed-wing nature of the platform, even when equipped solely with an RGB imaging payload, enables a broad range of potential applications. These include large-area surveying, environmental monitoring, infrastructure inspection, and situational awareness tasks where endurance and coverage are critical. Beyond data collection, the project aims to incorporate onboard and post-processing image analysis capabilities, directly supporting research and teaching activities in aircraft systems and flight, image processing and computer vision, and machine learning within the university. This integration provides a practical testbed for the development, validation, and deployment of advanced onboard decision making, including computer-vision-assisted perception and future autonomous mission logic, in a real-world aerial sensing context.

The project is structured and delivered through the QUT Aerospace Society, empowering the club to actively support students in applying skills acquired throughout their studies and to extend these capabilities through a practical, real-world engineering initiative aligned with QUT's applied learning focus. By providing a student-led platform for hands-on development, the project enables the direct application of core engineering concepts across aerospace, electrical, robotics, mechanical, and software disciplines throughout the project lifecycle.

Furthermore, the project positions the club as a conduit between students and industry, fostering engagement with relevant industry partners and exposing participants to professional engineering environments. This engagement supports network development and the establishment of meaningful connections with industry stakeholders. Through the club's operation of the platform, students will also gain practical exposure to the regulatory frameworks governing unmanned aircraft systems, as well as the operational practices, safety considerations, and compliance requirements expected when operating in a professional and commercial context.

The avionics and software architecture is intentionally platform-agnostic, allowing lessons learned to transfer to future QUTAS fixed-wing and multirotor platforms.

QUTAS intends to leverage the resources already at its disposal and to continue developing and improving projects to which it has already committed. Given the significant time and financial investment already made, and the substantial benefits this project provides to both the club and the university, QUTAS considers this project to be a valuable and justified continuation.

Timeline:

In the coming months, the project aims to incorporate a redundant secondary control link using a conventional RC controller and onboard receiver. This approach is common practice in UAS architectures and provides an additional layer of safety and resilience in the command and control system. The preferred protocol for this link is ExpressLRS (ELRS), a long-range, LoRa-based RC protocol widely used in advanced FPV and research platforms. This would provide a robust secondary control channel while also allowing the club to standardise on a single RC system across both fixed-wing and multirotor projects.

In the mid-term, the project seeks to fine-tune the parameters onboard the flight controller to enable the use of more advanced functionality, including robust failsafe procedures. During this phase, a deeper working understanding of ArduPilot and its more advanced features is expected to develop within the team. In parallel, it is anticipated that the project will establish stronger industry connections and that a suitable facility for regular flight testing will be identified.

In the longer term, the project seeks to integrate a companion computer to enable more advanced autonomous behaviours, along with a camera system to provide a pilot video feed and to serve as the primary means of acquiring RGB data for onboard processing or post-flight analysis. Beyond this, the platform could be further tailored and optimised, including the development of a university-designed and manufactured airframe and the incorporation of payload deployment capabilities. Additional peripheral sensors may also be integrated to enhance both situational awareness and sensing capability.

Phase	***Timeframe***	***Key Activities***
Near-term	0–3 months	Finalise avionics integration, establish redundant RC control link, conduct ground testing and initial flight testing.
Mid-term	3–6 months	Flight controller tuning, implementation of robust failsafe procedures, expansion of flight-testing activities, and strengthening of industry and testing facility engagement.
Long-term	6–12+ months	Integration of companion computer and camera payload, development of advanced autonomous and sensing capabilities, and platform optimisation for research and teaching use.

Table 1: Project timeline

Risks and mitigation

The primary risks associated with this project arise from the use of spinning propellers, unmanned flight operations, and lithium polymer batteries. Well established safety practices exist for each of these hazards, including safe motor handling during testing and operation, compliance with Civil Aviation Safety Authority (CASA) regulations for unmanned aircraft, and adherence to manufacturer safety data sheets for all components. Risk to people and property will be further mitigated through structured testing conducted within suitable, controlled facilities that provide adequate separation from bystanders and surrounding infrastructure. All flight operations will be undertaken by a suitably trained pilot holding the appropriate remote pilot licence, with insurance coverage provided through an organisation such as the Model Aeronautical Association of Australia (MAAA), ensuring regulatory compliance and appropriate protection in the event of an incident.

Budget

The project already has a strong foundation, including a suitable airframe, with a backup airframe currently in the possession of the club. This represents a prior investment by the club into the platform, and the intention of this proposal is to continue building on an established system rather than initiating a new project from scratch. At present, the system is equipped with a flight controller, a primary long-range communications link, and a basic sensor suite, including a GPS receiver and a low-grade pitot tube. While this configuration is sufficient for initial operation and testing, the avionics suite remains limited in capability. As such, the project seeks to upgrade the avionics stack to a more modern, cutting-edge configuration, enabling improved navigation robustness, reliability, and performance. These upgrades are critical to supporting the future development of advanced features envisioned for the project, including higher levels of autonomy, improved sensor fusion, and expanded sensing and communications capabilities.

1. RadioMaster GX12 Dual-Band Gemini-X Radio Controller Transmitter (~$350)

This is RadioMaster's latest transmitter and features built-in Gemini-X, ExpressLRS's dual-band transmission system that simultaneously operates on 2.4 GHz and 900 MHz. ExpressLRS has emerged as the de facto industry standard long-range RC protocol across both advanced FPV and research-oriented unmanned aircraft platforms, owing to its low latency, high reliability, and strong interference resilience. The dual-band Gemini-X implementation further improves link robustness by providing redundancy across frequency bands, outperforming traditional single-band systems.

At present, the club does not possess any ExpressLRS-compatible transmitters. Investing in a modern, high-capability ELRS radio at this stage establishes a common RC standard for future fixed-wing and multirotor projects, reducing fragmentation, simplifying training and support, and avoiding incremental upgrades across multiple platforms. While the initial cost is higher than entry-level alternatives, this approach represents a deliberate, forward-looking investment that improves reliability, scalability, and long-term value for the club.

2. RadioMaster XR4 Gemini Xrossband Dual Band ExpressLRS Receiver (~$50)

The XR4 offers low-latency, long-range performance and integrates cleanly with modern flight controllers, making it well suited to fixed-wing platforms. Its use supports a robust failsafe architecture while enabling the club to standardise on ExpressLRS receivers for future aircraft, improving reliability, consistency, and scalability across projects.

3. U-blox GEP-M10-DQ (~$43) or ZED-F9P Positioning Module (~$220)

The system currently uses a u-blox NEO-7 GPS module, which provides basic positioning but is limited in capability, particularly as it lacks an integrated magnetometer or barometer. Combined with the Matek F405 Wing, which also has no onboard magnetometer, we therefore have no direct magnetometer data for our navigation solutions. While this setup is acceptable for basic waypoint navigation, it offers sub-optimal heading and altitude estimation and fewer tracked satellites than modern receivers. Upgrading to a u-blox M10-series receiver would provide multi-constellation support (GPS, GLONASS, Galileo, BeiDou, etc.) with faster fixes and more robust satellite coverage, improving the input to our Kalman filter and yielding more stable flight with relatively modest cost. In contrast, a u-blox F9P module, though significantly more expensive, adds multi-band GNSS reception and advanced features such as reduced multipath errors and the potential for centimetre-level RTK corrections, resulting in much higher positioning accuracy and resilience in challenging environments. The F9P's enhanced performance can translate into tighter navigation, better heading and altitude estimation when fused with other sensors, and higher confidence in autonomous operations. Therefore, while the M10 represents a solid mid-range improvement, the F9P justifies its higher price when the mission demands precision, reliability, and scalability beyond standard receiver performance.

4. IMX335 5MP USB Camera (~$64)

A Suitable camera is required to capture and record images and video which can be stored locally for mission decision making and post processing. This camera breakout board provides a low cost and low form-factor solution capable of recording in 2K resolution and easily configurable with raspberry pi Linux platforms, making it an excellent choice for this application.

5. Turnigy 2200mAh 6S 60C LiPo Pack w/XT60 Connector (~$46)

A suitable onboard battery is required to supply stable electrical power to the propulsion system, avionics, flight controller, sensors, and communications hardware throughout all phases of flight, including high-load conditions such as launch, climb, and manoeuvring. The Turnigy 2200 mAh 6S 60C LiPo pack is well suited to this role for development and testing flights, as its 6S voltage aligns with common fixed-wing power systems while keeping current levels manageable. The high 60C discharge rating ensures the battery can comfortably deliver the transient high currents required without excessive voltage sag, supporting reliable motor and avionics performance. Its relatively modest capacity and weight also make it appropriate for early integration and flight testing, allowing the airframe to remain within safe mass limits while still providing adequate flight time for commissioning and evaluation activities.

6. Miscellaneous & Sundries (~$300)

This allocation will be used to cover unforeseen costs and sundry items required throughout the project, including consumables such as tape, paint, adhesives, fasteners, wiring, connectors, and other minor hardware. These items, while individually low in cost, are essential for airframe finishing, component installation, repairs, and iterative modifications during testing and integration, and providing a contingency for such expenses helps ensure the project can progress efficiently without delays due to small but necessary purchases.

Item	Online Shopping Link	Cost
RadioMaster GX12 Dual-Band Gemini-X Radio Controller Transmitter	Radio Controller Link	$350
RadioMaster XR4 Gemini Xrossband Dual Band ExpressLRS ELRS Receiver	Radio Receiver Link	$50
GPS Module	GPS Module Link	$220
Camera	Camera Link	$64
Battery	Battery Link	$46
Miscellaneous and Sundries	N/A	$300
	Total Cost	$1,030

Table 2: Itemised Budget with external Shopping Links